Problem A

Maria and the Longest Chain

TIme Limit: 2 Seconds

Input: Standard Input Output: Standard Output

Maria is very fond of chains. She has reached a large jewelry hall. There are large showcases full of beads of different sizes. She wants to have a long chain but at a time she can access only one showcase. Though all the beads are visible, she can access only the first row. She can access the adjacent next row, where the bead size is the same with the condition that she can access only the beads on the right of the same size bead in the next row. To build the longest chain Maria has to make the right choices. Help Maria to build the longest chain. The chain follows these rules:

1. It starts at the first element of list 1.
2. Within a list, you can move only to the right, one element at a time.
3. At any element with value x in list i, you may connect to an occurrence of the same value x in list $i + 1$. After connecting, you continue moving right from that position in the next list.
4. When connecting two lists, the shared value is counted only once.
5. You may stop at any position in any list. However, to move to the next list, you must connect through a common value.
6. If a later list cannot be reached, you may finish the chain in an earlier list.

The length of a chain is the number of elements it contains, counting each connection value only once. Find the maximum possible chain length.

Input Format

- The first line contains an integer N, the number of lists.
- Each of the next N lines contains an integer k_i, followed by k_i positive integers representing list i, in left-to-right order.

Output Format Print one integer: the maximum possible length of a chain.

Constraints $1 \le N \le 10^5$ $1 \le \sum k_i \le 2 \times 10^5$ $1 \le a_{i,j} \le 10^9$

Sample Cases:

Input	4 5 1 3 5 7 9 6 2 5 6 7 8 10 5 4 6 7 11 12 4 7 10 12 13	2 3 1 2 3 3 3 4 5	2 3 3 1 2 3 1 2 3	3 3 5 5 5 2 5 5 4 5 5 5 5	3 2 1 2 2 2 3 2 4 5
Output	8	5	4	7	3

Explanation 1. One of the longest chains is: 1 3 5 | 6 | 7 11 12 | 13. The chain starts in list 1, connects to list 2 at 5, then to list 3 at 6, and finally to list 4 at 12. The shared connection values are counted only once. The resulting chain has 8 elements.

2. The chain is 1 2 3 | 4 5. The lists connect at 3, giving a total length of 5.

3. The chain is 3 1 2 | 3. It connects at 2 in the first list to 2 in the second list, then continues right to 3. The shared 2 is counted once, so the chain contains 3, 1, 2, 3, with a total length of 4.